

Multiple Choice

1. **Answer:** B

Options: A) Volt; B) Weber; C) Watt; D) Ohm; E) Joule

Note: Correct option: B - Weber

2. **Answer:** A

Options: A) $\sinh x$; B) $e^x - e^{-x}$; C) $e^x + x$; D) $x^2 e^x$; E) $\cosh x$

Note: Correct option: A - $\sinh x$

3. **Answer:** C

Options: A) Low.; B) Very low.; C) High.; D) Almost zero.

Note: Correct option: C - High.

4. **Answer:** A

Options: A) It does not load the circuit at all.; B) It uses a potentiometer, which is more accurate than a voltmeter.; C) It uses a more precise measuring tool than a voltmeter.; D) It uses centre zero galvanometer instead of voltmeter.; E) It uses an ammeter instead of a voltmeter.

Note: Correct option: A - It does not load the circuit at all.

5. **Answer:** A

Options: A) $T_B = (a / 2 bR)$; B) $T_B = (R / ab)$; C) $T_B = (2 a / bR)$; D) $T_B = (aR^2 / b)$; E) $T_B = (bR / a)$

Note: Correct option: A - $T_B = (a / 2 bR)$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $k / (s^2 + k^2)$ '.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Initial value is 2, slope is -4'.

Long Form Response

11. **Answer:** 4.73. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Average mass velocity is 0 and molar average velocity is - 0.42 i m/sec. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key